From Peas to Probability: A Statistical Test of Mendelian Ratios

Mathematics Applications and Interpretation Higher Level

Internal Assessment

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How can probability be used to predict the inheritance of specific genetic traits from parents to their offspring?

Introduction

As someone who is a twin, I have always been fascinated by how two people can share nearly an identical set of DNA, yet differ at the same time. It's crazy to believe how it starts with a fertilized egg; people are created and their lives are determined, at least genetically. An error can lead to a change in fate. I always wondered how or why this happens, how people end up the way they are, or how they came to be. This fascination and curiosity has pushed me to explore how mathematics can explain biological variation. I selected this topic as it specifically demonstrates how something as broad as probability can explain something as fundamental to humans like inheritance.

Genetics is known as the study of how physical traits like eye color and hair color and inborn traits like the likelihood of inheriting a disease travel from one generation to the next. Genotypes are the genetic make up of an organism while phenotypes are observable or physical traits that result from the genotype. Genetics is due to how each parent contributes one set of alleles—different versions of the gene— to their offspring, which combines in order to determine an individual's genotype and phenotype. Dominant alleles on a punnett square are displayed as a capital letter, while recessive a lowercase letter. This process in which genotypes and phenotypes are inherited can be modeled mathematically through probability. Probability is the likelihood that an event will occur. One of the most common tools that is used to represent genetic probabilities is the Punnett square—a visual representation of the possible combinations of parental alleles as well as the corresponding probabilities of producing offspring with specific phenotypes.

The bridge between both genetics and mathematics is not only theoretical as the laws of probability are the base for the foundation of Mendelian genetics—the study of how specific

traits are passed from parents to offspring through genes. This form of genetics is a model established by Gregor Mendel who aimed to explain the inheritance of traits. Mendel's experiments originally incorporated pea plants that demonstrated traits that follow a predictable ratio (3:1 ratio of dominant to recessive traits on a monohybrid cross—genetic experiment that analyzes the inheritance of a single trait). Independent assortment is one of the fundamental principles of inheritance, discovered by Mendel. It states that alleles for different genes are disturbed into gametes independently from one another only as long as the genes are located on either different chromosomes or lay far apart on the same chromosome. This can be modeled through combinatorial models and binomial distributions, as these principles allow geneticist and biologists to anticipate the probability of phenotypes or traits that will appear in populations, or in some cases test if real data fits the theoretical expectations with tools like the chi-square test and t-tests.

The aim of this investigation is to model the genetic trend of inheritance using the laws of probability and algebraic methods like the binomial theorem, expected value, as well as conditional probability. Through the application of both binomial and normal distributions, I aim to investigate how likely it is for certain traits to appear within various offspring and interfere whether or not real or simulated data aligns with the theoretical genetic ratios. In addition, I will apply chi-square tests to evaluate the goodness of fit between both the experimental and the expected outcomes in monohybrid genetic crosses. My hypothesis is that these results will reflect the classical Mendelian ratios and demonstrate the probabilistic reasoning and accurately predict genetic variation.

Monohybrid:

H_0: Observed phenotype counts follow the expected Mendelian proportions for this cross

$H_{1:,}$ Observed phenotype count do **not** follow the expected proportions

Dihybrid:

H_0: Observed phenotype counts follow the expected 4-category Mendelian proportions

H_1: Observed phenotype counts do **not** follow the expected 4-category Mendelian proportions

Mathematical Tools and Data Sources

Monohybrid Crosses

Monohybrid crosses are genetic crosses that examine the inheritance of one trait controlled by two alleles.

	D	r
r	Dr	rr
r	Dr	rr

Dihybrid Crosses

Dihybrid crosses examine the inheritance of two traits simultaneously, where each is controlled by its own pair of alleles

	DR	Dr	dR	dr
DR	DDRR	DDRr	DdRR	DdRr
Dr	DDRr	DDrr	DdRr	Ddrr
dR	DdRR	DdRr	ddRR	ddRr
dr	DdRr	Ddrr	ddRr	ddrr

This will provide the visual framework in order to list all possible allele matchups. For dihybrid crosses with two traits, I will use a 4x4 grid in order to represent the different combinations of alleles from both parents.

Data Collection and Modeling

In order to gather realistic data:

- I will simulate various inheritance patterns using spreadsheets or digital Punnett square generators, allowing me to model probabilities for multiple offspring and test how observed results align with theoretical predictions.
- I will compare these simulated datasets to known experimental ratios (3:1 for monohybrid crosses or 9:3:3:1 for dihybrid crosses).

Data

Monohybrid Crosses Table

Round (R) is the dominant allele for seed shape, while wrinkled (r) is recessive. While in cotyledon color, yellow (Y) is the dominant allele and green (y) is recessive.

			Expected						**Observed**	
			Ratio							
Trait	**Parents**	**n**	**Genotype (1:2:1)**			**Phenotype (3:1)**		**Dominant**	**Recessive**	**Dominant**	**Recessive**
			AA	**Aa**	**aa**	**Dom.**	**Rec.**				

Seed Form	Round × Wrinkled	7,324	.25	.50	.25	.75	.25	5,493.00	1,831.00	5,474	1,850
Cotyledon Color	Yellow × Green	8,023	.25	.50	.25	.75	.25	6,017.25	2,005.75	6,022	2,001

Dyhybrid Crosses Table - Seed Form + Cotyledon Color

Round (R) is the dominant allele for seed shape, while wrinkled (r) is recessive. For cotyledon color, yellow (Y) is dominant and green (y) is recessive. As a result, the phenotypes are classified as: RY represents round seeds with yellow cotyledons, Ry represents round seeds with green cotyledons, rY represents wrinkled seeds with yellow cotyledons, and ry represents wrinkled seeds with green cotyledons.

	F1			**F2**	**Mendelian**		**(n)**	**Count**	
Parents	**Phenotype**	**Genotype**	**Gametes**	**Phenotype**	**Ratio**	**Fraction**	**Total**	**E**	**O**
Round Yellow seeds × Wrinkled Green seeds	All round, yellow seeds	RrYy	RY, Ry, rY, ry	Round Yellow	9/16	0.5625	556	312.75	315
Round Yellow seeds × Wrinkled Green seeds	All round, yellow seeds	RrYy	RY, Ry, rY, ry	Round Green	3/16	0.1875	556	104.25	108

Round Yellow seeds × Wrinkled Green seeds	All round, yellow seeds	RrYy	RY, Ry, rY, ry	Wrinkled Yellow	3/16	0.1875	556	104.25	101
Round Yellow seeds × Wrinkled Green seeds	All round, yellow seeds	RrYy	RY, Ry, rY, ry	Wrinkled Green	1/16	0.0625	556	34.75	32

Statistical Analysis

Monohybrid: Seed Form (Round × Wrinkled)

n = 7,324 offspring

Phenotypic expectation:

Dominant : Recessive = 3 : 1

p(dominant) = 3/4 = 0.75 p(recessive) = 1/4 = 0.25

Chi-square goodness-of-fit (3:1)

Formula: $X^2 = \sum \frac{(O-E)^2}{E}$

1) **Dominant**:

$O_D - E_D = (5{,}474 - 5{,}493)^2 = (-19)^2 / 5493 = 361/5{,}493 \approx 0.06573$

I will plug in my observed dominant value of 5,474 and compare it to the expected dominant value of 5493.

2) **Recessive**:

$O_R - E_R = (1{,}850 - 1{,}831)^2 = 19^2 / 1831 = 361 / 1{,}831 \approx 0.19720$

I will plug in my observed recessive value of 1.850 and compare it to the expected recessive value of 1,831.

Sum $\chi^2 = 0.06573 + 0.19720 = 0.26293$

Df: $2 - 1 = 1$ p-value: $P(\chi^2_1 \geq 0.26293) \approx 0.6081$

Since p-value (0.61) > α (0.05), we fail to reject the null hypothesis. The observed counts are consistent with a 3:1 ratio.

Normal approximation to the binomial

Model: $X \sim B(n = 7{,}324, p = 0.75)$

Observed X = 5,474 - 5493 = −19 **z** = $-19 / 37.0699 \approx -0.5127$

p-value from $z \approx 0.51$ (matches chi-square conclusion).

Likelihood of Genetics

P = probability of Mendelian genetics (dominant)

H_0: p=0.75 (Mendelian expectation) H_1: p≠0.75

Likelihood function: $L(p \mid x)=\binom{n}{x}p^x(1-p)^{n-x}$

$L(p \mid x)=\binom{7324}{5474}(0.75)^{5474}(1-0.75)^{1850}$

Maximum Likelihood Estimator: $\hat{P} = \frac{n}{x} = \frac{5474}{7324} \approx 0.7474$

Likelihood Ratio Test Statistic

$\Lambda=\frac{L(p=0.75)}{L(\hat{P})} = -2\ln(\Lambda) \sim x_1^2 = 0.26293$

This value matches the chi-square statistic previously calculated. Because p≈0.61 > 0.05 we fail to reject H_0 as the likelihood-based hypothesis test confirms that the observed data are consistent with a Mendelian 3:1 ratio.

95% confidence interval for dominant proportion

Observed proportion: $\hat{p}$ = 5,474 / 7,324 ≈ 0.7474057892

Standard Error (SE) = $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \approx \sqrt{(0.7474057892 \times 0.2525942108 / 7{,}324)} \approx$ 0.0013407

95% CI = $\hat{p} \pm 1.96 \times SE \approx 0.7474057892 \pm 1.96 \times 0.0013407 = (0.7374546723, 0.7573569061)$

0.75 lies comfortably inside the Confidence Interval of 95%, therefore the observed data is consistent with Mendelian expectation for monohybrid rosses

Monohybrid: Cotyledon Color (Yellow × Green)

n = 8,023 offspring

Phenotypic expectation: Dominant : Recessive = 3 : 1

I took 0.75 which is the probability that dominant alleles will be produced and I multiplied it by the total number of offspring (8,023) which produced my expected dominant value of 6,017.25 and multiplied 0.25 (probability of recessive alles being produced) by the total number of offspring, resulting in the value of 2005.75. The observed dominant and recessive values based off of Mendel's Data is 6,022 and 2,001.

Chi-square goodness-of-fit (3:1)

Sum $\chi^2 = 0.0149985458 \approx 0.01500$

df = 2-1= 1 p-value: ≈ 0.9025

Since P-val (0.90) > than α (0.05), we fail to reject the null hypothesis as the data proves to be extremely consistent with a 3:1 ratio.

Normal approximation to the binomial

Model: X ~ B(n = 8,023, p = 0.75)

Observed X = 4.75 **z** ≈ 0.1225

p ≈ 0.9025 (Value matches the chi-square).

Likelihood of Genetics

P = probability of Mendelian genetics (dominant)

H_0: p=0.75 (Mendelian expectation) H_1: p≠0.75

Maximum Likelihood Estimator: $\hat{P} = \frac{6022}{8023} \approx 0.7506$

Likelihood Ratio Test Statistic: $-2\ln(\Lambda) = 0.01500 \sim x_1^2$

Since $p \approx 0.90 > 0.05$, we fail to reject H_0. The likelihood under the Mendelian model is nearly identical to the maximum likelihood, which indicates very strong agreement with the 3:1 ratio expectation.

95% confidence interval for dominant proportion

Observed proportion: 6,022 / 8,023 ≈ 0.7505920479 **SE** ≈ 0.0019694

95% CI ≈ 0.7505920479 ± 1.96 × 0.0019694 → (0.7411243410, 0.7600597547)

0.75 lies comfortably inside the Confidence Interval of 95%, therefore the observed data is consistent with Mendelian expectation for monohybrid crosses

Dihybrid: Seed Form + Cotyledon Color (RrYy × RrYy)

Dataset: n = 556 offspring

Phenotypic fractions for two independent traits: There are a total of 16 different allele combinations that can be produced in a dihybrid cross.

Likelihood of Genetics

H_0: $(p_{RY}, p_{Ry}, p_{rY}, p_{ry}) = (\frac{9}{16}, \frac{3}{16}, \frac{3}{16}\frac{1}{16})$ H_1: At least one phenotypic probability differs

(X_1,X_2,X_3,X_4)~Multinomial(n=556, $\hat{P}$)

Likelihood function: $L(p_1, p_2, p_3, p_4)= 556! / (O_1! \cdot O_2! \cdot O_3! \cdot O_4!)\times p_1^{O_1} \cdot p_2^{O_2} \cdot p_3^{O_3} \cdot p_4^{O_4}$

$-2 \ln(\Lambda)$ follows a χ^2 distribution with 3 degrees of freedom

$-2 \ln(\Lambda) \approx 0.470$ p-value ≈ 0.93

Since p-value > 0.05, we fail to reject the null hypothesis. A likelihood ratio test was used to formally compare the Mendelian multinomial model with the maximum likelihood estimates, showcasing the theoretical equivalence with the chi-square goodness-of-fit test.

- **Round Yellow (RY):**

9/16 = 0.5625 Expected Count: $E_{RY} = 0.5625 \times 556 = 312.75$

- **Round Green (Ry):**

3/16 = 0.1875 Expected Count: $E_{Ry} = 0.1875 \times 556 = 104.25$

- **Wrinkled Yellow (rY):**

3/16 = 0.1875 Expected Count: E_{rY} = 104.25

• Wrinkled Green (ry):

1/16 = 0.0625 Expected Count: E_{ry} = 34.75

Observed counts:

• O_{RY} = 315 • O_{Ry} = 108 • O_{rY} = 101 • O_{ry} = 32

Chi-square goodness-of-fit (9:3:3:1) = $\chi^2 = \Sigma \frac{(Oi - Ei)^2}{Ei}$

1) Round Yellow (RY):

$O - E = (315 - 312.75)^2 = (2.25)^2 / 312.75 = 5.0625 / 312.75 \approx 0.01619$

2) Round Green (Ry):

$O - E = (108 - 104.25)^2 = (3.75)^2 / 104.25 = 14.0625 / 104.25 \approx 0.13495$

3) Wrinkled Yellow (rY):

$O - E = (101 - 104.25)^2 = (-3.25)^2 / 104.25 = 10.5625 / 104.25 \approx 0.10134$

4) Wrinkled Green (ry):

$O - E = (32 - 34.75)^2 = (-2.75)^2 / 34.75 = 7.5625 / 34.75 \approx 0.21755$

Sum $\chi^2 \approx 0.01619 + 0.13495 + 0.10134 + 0.21755 = 0.47002$ (rounded)

Df: 4 − 1 = 3 **p-value:** $P(\chi^2_3 \geq 0.47002) \approx 0.9254$

Since p-value (0.93) > $\alpha(0.05)$ we fail to reject the null hypothesis as the data are consistent with 9:3:3:1 expectation of independent assortment

Standardized residuals

Standardized residual = (O − E) / $\sqrt{E}$.

• RY: $(315 - 312.75) / \sqrt{312.75} \approx 0.1272$

• rY: $(101 - 104.25) / \sqrt{104.25} \approx -0.3183$

• Ry: $(108 - 104.25) / \sqrt{104.25} \approx 0.3673$

• ry: $(32 - 34.75) / \sqrt{34.75} \approx -0.4665$

There are no unusually large cell residuals.

Observed proportions & 95% CIs = $\hat{p} = \frac{O}{n}$

Standard Error (SE) = $\sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}}$ **95% CI** = $\hat{p} \pm 1.96 \cdot SE$

Round Yellow (RY)

$\hat{p} = \frac{O}{n}$ = 315/556 = 0.566546762590

$\hat{p}(1 - \hat{p})$ = 0.566546762590(1 − 0.566546762590) = 0.245571528389 / 556 = 0.000441675411

SE = $\sqrt{0.000441675411}$ = 0.021016075056

95% CI = 0.566546762590 $\pm$ 1.96 * 0.021016075056 = (0.525355, 0.607738)

Round Green (Ry)

$\hat{p} = \frac{O}{n}$ = 108/556 = 0.194244604317

$\hat{p}(1 - \hat{p})$ = 0.194244604317 (1 − 0.194244604317) = 0.156513638010 / 556 = 0.000281499349

SE = $\sqrt{0.000281499349}$ = 0.016777942333

95% CI = 0.194244604317 $\pm$ 1.96 * 0.016777942333= (0.161360, 0.227129)

Wrinkled Yellow (rY)

$\hat{p} = \frac{O}{n}$ = 101/556 = 0.181654676259

$\hat{p}(1 - \hat{p})$ = 0.181654676259 × (1 − 0.181654676259) = 0.148656254852 / 556 = 0.000267367365

SE = $\sqrt{0.000267367365}$= 0.016351371956

95% CI = 0.181654676259 $\pm$ 1.96 * 0.016351371956 = (0.149606, 0.213703)

Wrinkled Green (ry)

$\hat{p} = \frac{O}{n}$= 32/556 = 0.057553956835

$\hat{p}(1 - \hat{p})$ = 0.057553956835 × (1 − 0.057553956835) = 0.054241498887/ 556 = 0.000097556653

SE = $\sqrt{0.000097556653}$ = 0.009877077132

95% CI = 0.057553956835 $\pm$1.96 * 0.009877077132(0.038195, 0.076913)

Results and Analysis

The pea plant data which was applied to the mathematical modeling provides strong evidence that Mendelian genetics is accurate and predicts the distribution of traits in large populations. With Monohybid crosses, the expected genotype ratio of 1:2:1 and the expected phenotype ratio were calculated using binomial probabilities and Punnett squares. In addition, the observed values for both seed forms (Round vs. Wrinkled) and cotyledon color (Yellow vs. Green) did closely match with their expected values (observed 5493 dominant vs. 1831 recessive) to (expected 5474 dominant vs. 1850 recessive). The chi-square calculations produced values below the critical value, which supports that random genetic variation can explain the tiny differences which are seen between observed and expected counts.

With dihybrid crosses, the analysis and results revealed the typical 9:3:3:1 phenotype ratio which stems from an independent assortment of alleles. These expected counts were calculated using the rational probabilities of dihybrid mendelian genetics (9/16, 3/16, 3/16, and 1/16) which were multiplied by the sample size of 556. The observed values (315, 108, 1101, 32) were very close to the predicted values of (312.75, 104.25, 104.25, and 34.75). In addition, the chi-square test confirmed that these deviations fall within the range of typical biological variation.

The binomial distribution and the normal approximation tests exemplified how the variance shrinks enough to the point where theoretical and real world results almost overlap. This strongly

supports Mendel's claim of how genetics fundamentally follows predictable patterns. Each mathematical method with punnett square reasoning and probability and statistical testing concluded the same conclusion: mendelian genetics is highly predictive and is mathematically consistent with large sample sizes.

With both monohybrid crosses (Seed Form: Round X Wrinkled) (Cotyledon Color: Yellow X green) and the dihybrid crosses (Round Yellow X Wrinkled Green), the observed data demonstrates Mendelian predictions for genotypic and phenotypic ratios. In the monohybrid cross with the seed form, the observed data with the 3:1 ratios is close to the expected values. The chi-squared value (0.26294) gives the p-value of about 0.61, which concludes that we fail to reject the null hypothesis as there is no significant deviation from Mendel's expected phenotypic ratio . In the second monohybrid cross with cotyledon color, the observed data is even close to the expected predictions as the chi-squared value is 0.015, giving the p-value of about 0.9, indicating that the data almost perfectly matches the Mendelian predicted ratios. In addition, in the dihybrid cross, the expected phenotypic ratio is 9:3:3:1 on proportion to independent assortment of the two parent traits. This demonstrated as the chi-squared value of 0.46002 gives the p-value of (0.9254). This indicates strong alignment with independent assortment, making these datasets and results reflect the theoretical Mendelian expected ratios for both monohybrid crosses and dihybrid crosses.

I learned that observed numbers can align closely with expected values and do not have to be a perfect match; they can instead reflect a similar pattern. While small deviations can occur due to different natural causes, statistical tests can prove these deviations to be insignificant.

Chi-squared tests can play a large role in testing genetic hypotheses as they allow you to

determine whether the data aligns with the predicted ratios. It also indicated whether to accept or reject the null hypothesis.

Limitations

Biological Assumptions

One limitation is that many traits are influenced by multiple genes, and alleles don't follow standard dominant/recessive patterns, In real biology, many traits always exhibit incomplete dominance, codominance, and polygenic inheritance, however, Mendelian models assume that traits are controlled by a single gene with complete dominance, which limits the generalizability of findings to traits that only follow Mendelian rules. Since Mendelian genetics assume that traits are controlled by one gene with complete dominance, that may not prove to be true with multiple samples in real organisms.

Sampling Constraints

An additional limitation is that the data utilized is from a controlled and historical dataset, rather than from self-collected samples. Since it was already collected, I have to rely on the accuracy of outside sources, so if there were environmental factors or experimental errors in the original study, I have no way of knowing.

Environmental variation

Another limitation is that although the sample sizes were vaguely large, different environmental factors like temperature, soil quality and others, could influence germination rates or survival, which in turn could potentially skew observed counts. These are external factors that cannot be accounted for mathematically.

Extension

One way that this experiment could be extended is to explore trihybrid crosses and polygenic traits, which would result with more specific probability. A trihybrid cross involves three independently assorting genes, which result in 8 potential gametes per parent. The F_2 phenotypic ratio is 27:9:9:9:3:3:3:1, where each additional trait leads to the total number of possible outcomes to increase exponentially. This in turn makes probability predictions more precise. However, this results in the data to be more sensitive to sampling variations.

In addition, using more advanced statistical tests or a more visual regression model could be used to visually compare the expected and observed values. It could demonstrate that these mathematical tools can be applied to more complex real-world genetics and not be restricted to Mendelian genetics. One approach for this could be the implementation of maximum likelihood estimation which is used to estimate recombination and allele probabilities that are directly observed from offspring counts. This allows parameters like recombination frequency and dominance probability to be inferred from the data itself and compared directly to theoretical values.

An additional extension is to compare different sample sizes and see how the sample size affects the accuracy of the predicted ratios. I could calculate smaller groups of offspring and see how much it varies compared to the larger sample sizes. This could account for why larger populations often match theoretical ratios more closely and while small experiments often have a large variation in error, even when the genetics are correct. This tests the reliability of predicted

Mendelian ratios, tests genetic predictions like 3:1 and 9:3:3:1 and represents long-run probabilities. However, due to proven data, observed ratios depend heavily on the total number of offspring sampled. Using a multinomial model for dihybrid and trihybrid crosses and a binomial model for monohybrid crosses is where one could analyze how variability decreases as sample sizes increase. This could impact the sample size even more as the offspring is divided into many phenotype categories in dihybrid and trihybrid crosses.

Conclusion

The results of this investigation closely matched Mendelian predictions of both the monohybrid cross and the dihybrid cross. The statistical analysis which utilized chi-squared tests demonstrated no significant differences between the observed and expected ratios of the pea plant data, which indicates that rather than non-Mendelian inheritance, the small deviations were due to random sampling variation. This investigation also indicated that larger sample sizes do produce more stable ratios and smaller confidence intervals, which in turn significantly improve the accuracy of genetic and allele predictions. Overall, the data does support Gregor Mendel's law of independent assortment and dominance as it demonstrates the significant value of statistical analysis in interpreting genetic results.

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